

A380X EMER CANC pushbutton

The ECP EMER CANC pushbutton of the A380 FCOM: warnings silenced, cautions cancelled for the flight, CANCELLED CAUTION on the STATUS page, recall with RCL held 3 s

Branch	pr/28-a380x-fws-emer-canc
Aircraft	A380X (FWS, EWD, SD)
Commits	4377b066a feat(a380x/fws): EMER CANC pushbutton
Base	upstream master
Status	CI checks pass; merged into develop (ca3644748); built and deployed; not tested in the sim yet (the FWS does not run in the browser harness).

Purpose

The EMER CANC pushbutton set L:A32NX_BTN_EMERCANC and nothing read it. The A380 FCOM gives it a precise role on the alert at the top of the EWD (DSC-31-40-20 P 5).

Specification (A380 FCOM DSC-31-40-20, DSC-31-40-10)

Alert	EMER CANC
Warning (level 3)	aural alert and MASTER WARN cancelled, procedure kept; back when the warning comes again
Caution (level 1 or 2)	cancelled for the rest of the flight with its procedure and MASTER CAUT; CANCELLED CAUTION section (white) on the STATUS MORE page
Always	all the aural alerts and audio indicators, playing and queued; the not-sensed procedures activated from ABN PROC
Nothing to cancel	memo EMERGENCY CANCEL ON for 3 s (time not given by the FCOM: as the RCL NORMAL)
RCL > 3 s	the cancelled cautions still active come back on the EWD
Reset	FWS power loss, 50 minutes after shutdown (as the other ECAM resets)

Code

- FwsEmerCancel.ts: cancelled cautions and silenced warnings (4 tests).
- FwsCore.ts: input buffer and pulse node as CLR / RCL; the cancelled cautions left out of the new warnings, the EWD procedures and the recall; FwsSoundManager.cancelAll(); memo 310000002; bus topic fws_cancelled_caution to the STATUS page.

PR description (from the PR draft)

From `pr-drafts/28-a380x-fws-emer-canc.md`, section "Summary of Changes" onwards.

Summary of Changes

The EMER CANC pushbutton of the ECP sets `L:A32NX_BTN_EMERCANC`, but nothing in the A380X FWS read it. It now works as the A380 FCOM describes it (DSC-31-40-20 P 5), on the alert at the top of the EWD:

- **Warning:** its aural alert and the MASTER WARN light are cancelled; the procedure stays on the EWD. When the warning goes away and comes back, the aural alert and MASTER WARN come back.
- **Caution:** cancelled for the rest of the flight, with its procedure and the MASTER CAUT light; it stays cancelled if the failure comes back. The cancelled cautions are listed in white in a new CANCELLED CAUTION section of the STATUS MORE page (DSC-31-40-10).
- **Always:** all the aural alerts and audio indicators (the one playing and the queued ones) are cancelled, and the not-sensed procedures activated from ABN PROC are deactivated.
- **Nothing to cancel:** the EWD shows the memo EMERGENCY CANCEL ON for 3 s (the FCOM gives no time; 3 s as the RCL NORMAL).
- **RCL pressed more than 3 s** brings back the cancelled cautions still active on the EWD (DSC-31-40-10).
- Reset with the other ECAM resets (FWS power loss, 50 minutes after shutdown).

Code: `FwsEmerCancel.ts` keeps the cancelled cautions and the silenced warnings (4 unit tests); `FwsCore.ts` reads the pushbutton through its input buffer and pulse node as CLR and RCL, and excludes the cancelled cautions from the new warnings, the EWD procedures and the recall; `FwsSoundManager.cancelAll()`; memo `310000002`; STATUS page section fed by the new bus topic `fws_cancelled_caution`.

Cockpit API Changes

None (the existing `L:A32NX_BTN_EMERCANC` is now read). New FWS bus topic `fws_cancelled_caution` (FWS -> SD).

Screenshots (if necessary)

TO ADD: STATUS page with the CANCELLED CAUTION section; EWD with EMERGENCY CANCEL ON.

References

- A380 FCOM DSC-31-40-20 P 5 (ECP, EMER CANC pb).
- A380 FCOM DSC-31-40-10 (CANCELLED CAUTION on the STATUS page, RCL pb pressed more than 3 s).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Cold and dark, then power: trigger a caution (e.g. a bleed or a pack off in flight) and press EMER CANC: MASTER CAUT off, the caution and its procedure removed; STATUS > MORE: CANCELLED CAUTION lists it. Trigger it again: it stays cancelled.
2. Hold RCL more than 3 s: the caution comes back on the EWD.
3. Trigger a warning with an aural alert (e.g. ENG FIRE TEST, or a failure from the flyPad) and press EMER CANC: aural and MASTER WARN off, procedure kept.
4. EMER CANC with no alert: EMERGENCY CANCEL ON for 3 s in the memo area.
5. `npx vitest run fbw-a380x/src/systems/systems-host/CpiomC/FlightWarningSystem/FwsEmerCancel.spec.ts` (4 tests).